MIL-F-50454A (MU)
19 March 1973
SUPERSEDING MIL-F-50454
15 August 1969

MILITARY SPECIFICATION

FUZE, GRENADE, M219E1

LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This Military Specification covers the loading, assembling and packing for one type of fuze designated as Fuze, Grenade, M219E1.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-Z-325	- Zinc Coating Electrodeposited, Requirement for
QQ-P-416	- Plating, Cadmium (Electrodeposited)

MILITARY

MIL-A-2550	- Ammunition and Special Weapons, General Specification for
MIL-I-45607	- Inspection Equipment, Acquisition, Maintenance and Disposition of

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)

FSC: 1330

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MIL-STD-109 - Quality Assurance Terms and Definitions

MIL-STD-331 - Jolt Test for Use in Production of Fuzes

MIL-STD-331 - Jumble Test for Use in Production of Fuzes

MIL-STD-331 - Transportation Vibration Test for Use in Production of Fuzes

MIL-STD-414 - Sampling Procedures and Tables for Inspection by Variables for Percent Defective

MIL-STD-1168 - Lot Numbering of Ammunition

MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

PURCHASE SPECIFICATION

MIL-D-14978 - Detonator, Stab, M55, Loading, Assembling and Packing

DRAWINGS

US ARMY MUNITIONS COMMAND

9207971 - Fuze, Grenade M219E1

PUBLICATIONS

ARMY

ET-9207971 - Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer).

2.2 Other publications.-The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

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CODE OF FEDERAL REGULATIONS

Title 49 - Transportation, Parts 171-179

(The Code of Federal Regulations is available from the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402. Orders should specify, "49 CFR 0-190 (latest revision)".)

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with the applicable drawings and specifications.

3.2 Parts.-The fuze parts and assemblies shall comply with all requirements specified on Drawing 9207971, and with all requirements specified in applicable specifications.

3.3 Arming of housing assembly.-The housing assembly, less detonator, shall arm before reaching 3201 revolutions per minute (RPM) when rotated about its longitudinal axis and tested as specified in 4.4.3.

3.4 Non-Arming of Housing Assembly.-The Housing Assembly, less detonator, shall not arm when spun on its axis to a spin velocity of 2400 revolutions per minute (RPM) and tested as specified in 4.4.4.

3.5 Verification of non-arming of housing assembly.-After the housing assembly, less detonator, has been subjected to the non-arming test in 4.4.4, it shall not arm or partially arm* if spun again on its axis to a spin velocity of 2300 revolutions per minute (rpm), when tested as specified in 4.4.5.

3.6 Staking of firing pin assembly.-The firing pin shall not separate or become loose from the firing pin spring when a minimum pull of two hundred (200) grams is applied as specified in 4.4.6.

3.7 Staking of detonator.-The detonator shall not move a distance greater than that specified on the applicable drawing, when tested and subjected to the force specified on that drawing.

3.8 X-Ray examination of sample fuze assemblies prior to jolt and jumble tests.-The fuze assembly samples for the jolt and jumble tests shall be x-rayed and shall not be armed or partially armed* when tested as specified in 4.2.2.1.7 and 4.3.3.7.

3.9 Jolt.-The fuze assembly shall withstand the jolt test without having any explosive elements explode, fuze come apart, fuze partially arm* or fuze arm when tested as specified in 4.4.9.

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3.10 Jumble.-The fuze assembly shall withstand the jumble test without having any explosive elements explode, fuze come apart, fuze partially arm* or fuze arm when tested as specified in 4.4.10.

3.11 Waterproofness.-The fuze assembly shall withstand the waterproofness test without emitting a continuous stream of air bubbles when tested as specified in 4.4.11.

3.12 Transportation vibration.-The fuze assembly shall withstand the transportation vibration test without having any explosive elements explode, fuze come apart, fuze partially arm* or fuze arm when tested as specified in 4.4.12.

*NOTE: "A "partially armed" unit is one in which two (2) or more adjacent lockweights are not locked in the detonator rotor. A lockweight is considered unlocked when the corner of the lockweight against the rotor is displaced a distance from the rotor to permit insertion of a .030 inch round pin." This will be determined on x-ray by the use of a .030" diameter circle or other suitable means. This definition applies only to radiographs of aluminum housing fuzes (P/N 9207971-1). Partially armed zinc housing fuzes (P/N 9207971-2) are distinguishable on a radiograph by marked lack of parallelism of the outer edge of two or more adjacent lockweights and the back edge of their respective pockets. This may be verified by placing on the fuze image a transparent overlay on which is drawn a slot at least 1/4" long by .045" in width, superimposing the outer side of the .045" slot with the back edge of the rotor housing pocket and observing if any portion of the lockweight image fails within the .045" slot.

3.13 Functioning of fuze assembly.-The fuze assembly shall function satisfactorily on contact with the impact medium. Functioning will be evidenced by initiation of the Detonator and Lead Cup when tested as specified in 4.4.13.

3.14 Workmanship.-The fuze parts and assemblies shall be fabricated and loaded in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, rust and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. Base metal visible, except that exposed by flaking, on the cover (P/N 9207967) caused by normal process machine operations is acceptable. All required markings shall be neat and sharply defined.

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3.15 Submission of results of contractor-conducted examinations and tests.-The contractor shall forward copies of the results of any of his examinations or tests to the designated address as required by the contract, the specification, or as requested by the Contracting Officer. (See 6.6)

3.16 First article testing.-This specification makes provisions for first article testing. Submission of first article quantity by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to the prescribed requirements. Reference shall be made to MIL-STD-109 in order to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Submission of product.-At the time the completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all the quality assurance provisions specified in this specification.
b. Drawing and specification number and date, together with identification and date of changes thereto.
c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.
d. Number of items in the lot.
e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

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4.1.2 Government verification.-Using the contractor's written quality assurance procedure, this detailed specification, the applicable drawings, and other contractural documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "Critical" be accepted solely on the basis of the contractor's records.

4.2 First Article Inspection.

4.2.1 Submission.-The contractor shall submit a first article quantity as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.2.2. (See 6.1) The first article sample shall consist of the following:

a. Ten (10) plated covers.
b. Two hundred fifty (250) housing assemblies, less detonator.
c. Eighty (80) firing pin assemblies.
d. Eighty (80) housing assemblies.
e. Five hundred seventy-four (574) fuze assemblies.
f. One hundred twenty-five (125) housing assemblies, with firing pin assembly and centerplate staked in place.
g. One hundred twenty-five (125) fuze assemblies, less lead cup assembly and housing seal.
h. Fifty (50) rotor and spring assemblies.
i. Fifty (50) rotor and housing assemblies.
j. Fifty (50) of each metal part.

NOTE: For die-cast parts, five (5) parts from each cavity or for stamped parts, five (5) parts from each tool shall be supplied in lieu of the fifty (50) of each metal part mentioned above.

All parts and completed assemblies submitted shall have been produced by the contractor (or furnished by a supplier) using the same production processes, procedures and equipment as will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same sources of supply as will be used in regular production. The sample shall be accompanied by certificates of conformance. A first article quantity, or portion thereof, as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process

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(including heat treatment procedures and equipment, material used, drawing specification or source of supply as to significantly affect product uniformity as determined by the Government. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample. A sample containing known defects will not be submitted unless specifically authorized by the Contracting Officer.

4.2.2 Inspections to be performed.-Assemblies and components will be subjected by the Government to any or all the examinations or tests specified in this specification and any or all requirements of the applicable drawings. In addition, any or all of the following examinations and tests will be performed.

4.2.2.1 Quantities for first article.-After inspection for classification of defects has been performed as specified in 4.3.2, the following quantities shall be subjected to the indicated tests.

4.2.2.1.1 Salt spray.-(See Dwg. 9207967) and Federal Specification QQ-Z-325 or QQ-P-416 as applicable). Ten (10) plated covers shall be subjected to this test in accordance with QQ-Z-325 or QQ-P-416 as applicable.

4.2.2.1.2 Arming of housing assembly (See 3.3).-One hundred twenty-five (125) housing assemblies, less detonator, shall be subjected to this test.

4.2.2.1.3 Non-arming of housing assembly (See 3.4).-After completion of the test specified in Paragraph 4.2.2.1.2 another one hundred twenty-five housing assemblies less detonators shall be subjected to this test."

4.2.2.1.4 Verification of non-arming of housing assembly (See 3.5).-After completion of the test specified in 4.2.2.1.3, the one hundred twenty-five (125) housing assemblies, less detonator, shall be subjected to this test.

4.2.2.1.5 Staking of firing pin assembly (See 3.6).-The eighty (80) firing pin assemblies shall be subjected to this test.

4.2.2.1.6 Staking of detonator (See 3.7).-The eighty (80) housing assemblies shall be subjected to this test.

4.2.2.1.7 X-Ray examination of jolt and jumble samples prior to testing.-Prior to subjecting the sample fuze assemblies to the tests specified in 4.2.2.1.8 and 4.2.2.1.9, the fuze assemblies shall be subjected to this test and meet the requirements of 3.8.

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4.2.2.1.8 Jolt (See 3.9).-Twelve (12) fuze assemblies shall be subjected to this test.

4.2.2.1.9 Jumble (See 3.10).-One hundred twenty-five (125) fuze assemblies shall be subjected to this test.

4.2.2.1.10 Waterproofness (See 3.11).-Fifty (50) fuze assemblies shall be subjected to this test.

4.2.2.1.11 Transportation vibration (See 3.12).-Five hundred (500) fuze assemblies shall be subjected to this test.

4.2.3 Rejection.-If any assembly or component fails to comply with any of the applicable requirements, the first article quantity shall be rejected. The Government reserves the right to terminate its inspection upon any failure of an assembly or component in the sample to comply with any of the stated requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new first article quantity or portion thereof. Until a first article quantity is accepted, the contractor is in no way authorized by the Government to resume regular production unless otherwise directed by the Contracting Officer.

4.2.4 Functioning of fuze assembly (See 3.13).-After completion of the transportation vibration test specified in 4.2.2.1.11, five hundred (500) fuze assemblies shall be subjected to this test.

4.2.4.1 Rejection.-If eleven (11) or more of the fuze assemblies fail to comply with the applicable requirement, the first article sample shall be rejected and the contractor shall take the necessary corrective action and submit a new first article sample as required.

4.2.5 First article inspection.-During first article inspection the provisions of 4.3, inspection provisions (for production), shall apply and all of the examinations and tests therein shall be performed except that the following tests shall be conducted on the special quantities specified, beginning with the first lot produced and continuing until three (3) consecutive lots have passed the acceptance criteria prescribed. Thereafter, sampling plans for these tests shall be in accordance with 4.3.3. (Production test sampling). All first article samples for these tests shall be selected by the Government inspector.

4.2.5.1 Transportation vibration (See 3.11).-Three hundred fifteen (315) fuze assemblies shall be subjected to this test. If one (1) or more fuze assemblies fail to comply with the transportation vibration test requirement, the lot shall be rejected.

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4.2.5.2 Functioning (See 3.12).-Three hundred fifteen (315) fuze assemblies subjected to the transportation vibration test in 4.2.5.1 shall be subjected to this test. If eleven (11) or more fuze assemblies fail to comply with the functional test requirements, the lot shall be rejected.

4.3 Inspection provisions (for production).-

4.3.1 Lot formation.-The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considerdd to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to process, specification, or drawing, not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot.

All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each inspection lot shall contain:

a. Detonators of one lot interfix number from one manufacturer.
b. Lead cup assemblies of one interfix number from one manufacturer.

4.3.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (See 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's to individual defects using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect.

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When this method is selected, it shall be applied to all of the major and minor defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.2 Fuze, Grenade M219E1, prior to assembly of housing seal and lead cup assembly (See Dwg. 9207971).-

Categories	Defects	Method of Inspection	Defect Code
Critical:			
1.	Detonator rotor not in safe position	Machine/Probe (See 4.4.14)	25001
Major:	None defined		
Minor:	None defined		

4.3.2.3 Fuze, Grenade M219E1, prior to assembly of housing seal (See Dwg. 9207971).-

Categories	Defects	Method of Inspection	Defect Code
Critical:	None defined		
Major	AQL 0.40 Percent		
101.	Lead cup assembly or charge missing	Visual	03001
Minor:	None defined		

4.3.2.4 Fuze, Grenade M219E1, prior to assembly of Asphalt Sealant, Lead Cup and Housing Seal (See Dwg. 9207971).-

Categories	Defects	Method of Inspection	Defect Code
Critical:			
1.	Absence of one (1) or more lockweight springs	Test	22001
Major:	None defined		
Minor:	None defined		

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4.3.2.5 Fuze, Grenade M219E1 (See Dwg. 9207971).-

Categories	Defects	Method of Inspection	Defect Code
Critical:	None defined		
Major	AQL 1.00 Percent		
101.	Outside diameter, max at lead cup end	Gage	04001
102.	Outside diameter, max on largest diameter	Gage	04002
103.	Outside diameter, max, on diameter of end opposite to lead cup	Gage	04003
104.	Length of shoulder, max	Gage	04004
105.	Length, max	Gage	04005
106.	Height, max of dimple	Gage	04006
Minor	AQL 1.00 Percent		
201.	Evidence of poor workmanship (See 3.13)	Visual	04007
202.	Marking misleading or uni-dentifiable	Visual	04008

4.3.2.6 Box, inner, prior to inserting supports (See Dwg 8882317 and 8882318).-

Categories	Defects	Method of Inspection	Defect Code
Critical:	None defined		
Major:	None defined		
Minor	AQL 0.65 Percent		
201.	Lower outer flaps not sealed	Visual	05002

4.3.2.7 Support, prior to inserting into inner box (See Dwg 8882317 and 8882318).-

Categories	Defects	Method of Inspection	Defect Code
Critical:	None defined		
Major	AQL 0.40 Percent		
101.	Support not closed	Visual	06003
102.	Absence of radius on lid corners	Visual	06004

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Minor	AQL 1.50 Percent		
201.	Support damaged	Visual	06005
202.	Incorrect number of fuzes	Visual	06006
203.	Pad missing	Visual	06007

4.3.2.8 Box, inner, prior to sealing outer flaps (See Dwg. 8882318).-

Categories	Defects	Method of Inspection	Defect Code
Major: None defined			
Minor	AQL 1.00 Percent		
201.	Incorrect number of supports	Visual	07001
202.	Supports improperly orientated	Visual	07002

4.3.2.9 Box, inner, sealed (See Dwg. 8882318).-

Categories	Defects	Method of Inspection	Defect Code
Critical: None defined			
Major	AQL 0.65 Percent		
101.	Box damaged	Visual	08001
102.	Box improperly sealed	Visual	08002
103.	Contents loose	Manual	08003

4.3.2.10 Box, outer sealed (See Dwg. 8882318).-

Categories	Defects	Method of Inspection	Defect Code
Critical: None defined			
Major	AQL 0.65 Percent		
101.	Box damaged	Visual	09001
102.	Box improperly sealed	Visual	09002
103.	Contents loose	Manual	09003
Minor	AQL 0.65 Percent		
201.	Marking misleading or unidentifiable	Visual	09004

4.3.2.11 Housing Assembly, Prior to Assembly of Detonator (See Dwg. 9215195).-

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Categories	Defects	Method of Inspection	Defect Code
Critical:			
	1. Detonator rotor not locked in unarmed position	Machine Probe (See 4.4.1)	02002
Major:	None defined		
Minor:	None defined		

4.3.3 Testing.

4.3.3.1 Salt spray (See Dwg. 9207967 and Federal Specification QQ-Z-325 or QQ-P-416, as applicable) Defect Code 10001.-A sample of four (4) plated covers shall be selected from each lot and tested as specified in 4.4.2. If one (1) or more covers fail to comply with the salt spray requirement in the applicable specification, the lot shall be rejected.

4.3.3.2 Arming of housing assembly (See 3.3) Defect Code 11001, Major Defect.-A sample of housing assemblies less detonator shall be selected from each lot in accordance with MIL-STD-105, Level II, AQL 0.65, Table II-A and subjected to this test. If the number of housing assemblies which fails to comply with the arming requirements exceeds the applicable number allowed by MIL-STD-105, the lot shall be rejected.

4.3.3.3 Non-arming of housing assembly (See 3.4) Critical Defect, Defect Code 12001).-Each housing assembly, less detonator, shall be tested as specified in 4.4.4, using equipment in accordance with 4.3.4. If any housing assembly fails to comply with the non-arming requirement, it shall be classified defective and removed from the lot.

4.3.3.4 Verification of non-arming of housing assembly (See 3.5), Critical Defect, Defect Code 13001).-After completion of the one hundred (100) percent inspection specified in 4.3.3.3, three hundred fifteen (315) housing assemblies, less detonator, shall be selected from each housing lot and subjected to the non-arming test. The test shall be performed as specified in 4.4.5, using equipment in accordance with 4.3.4. If one (1) or more housing assemblies fail to comply with the verification non-arming requirement, the lot shall be rejected.

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4.3.3.5 Staking of firing pin assembly (See 3.6) Defect Code 14001.-A sample of fifty (50) firing pin assemblies shall be selected from each lot and subjected to this test. The test shall be performed as specified in 4.4.6, using equipment in accordance with 4.3.4. If one (1) or more firing pin assemblies fail to comply with the applicable requirement, the lot shall be rejected.

4.3.3.6 Staking of Detonator (See 3.7) Defect Code 15001.- A sample of eighty (80) housing assemblies shall be selected from each lot and subjected to this test. The test shall be performed as specified in 4.4.7, using equipment in accordance with 4.3.4. If two (2) or more assemblies fail to comply with the applicable drawing requirement, the lot shall be rejected.

After ten (10) consecutive lots have been accepted according to the above sampling plan, the following double sampling plan will be used:

An initial sample of thirty-two (32) Housing assemblies will be selected from each lot and subjected to this test. If two (2) or more failures occur, the lot shall be rejected. If one (1) failure occurs, a second sample of ninety-six (96) housing assemblies shall be tested. If the total number of failures equals or exceeds two (2), the lot shall be rejected.

If two consecutive lots are rejected when the double sampling plan is in use, the contractor must re-qualify his process by passing ten (10) consecutive lots according to the single sample plan.

4.3.3.7 X-ray examination of sample fuze assemblies prior to jolt and jumble test (See 3.8) Critical Defect, Code 23001.-The samples selected for the tests specified in 4.3.3.9 and 4.3.3.10 shall be subjected to this test. If any assembly fails the requirement of 3.8, the lot shall be rejected. This test shall be performed as specified in 4.4.8.

4.3.3.8 Jolt (See 3.9) Critical Defect, Defect Code 16001.-A sample of twelve (12) fuze assemblies shall be selected from each lot and subjected to this test. Prior to testing, the sample shall be x-rayed as required by 3.8. The jolt test shall be performed as specified in 4.4.9 using equipment in accordance with 4.3.4. If one (1) or more fuze assemblies fail to comply with the Jolt test requirement, the lot shall be rejected.

4.3.3.9 Jumble (See 3.10) Critical Defect, Defect Code 17001.- A sample of one hundred twenty-five (125) fuze assemblies shall be selected from each lot and subjected to this test. Prior to testing, the sample shall be x-rayed as required by 3.8. The jumble test shall be performed as specified in 4.4.10, using equipment in accordance with 4.3.4. If one (1) or more fuze assemblies fail to comply with the jumble test requirement, the lot shall be rejected. If the detonator of one (1) or more assemblies fires while in the safe position, the lot shall be rejected. Major Defect, Defect Code 17002.

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4.3.3.10 Waterproofness (See 3.11) Defect Code 18001.-A sample of fifty (50) fuze assemblies shall be selected from each lot and subjected to this test. The test shall be performed as specified in 4.4.11, using equipment in accordance with 4.3.4. If in the original test, five (5) or more fuze assemblies fail to comply with the applicable requirement, the lot shall be rejected. If three (3) or four (4) fuzes fail to comply with the applicable requirement, a retest will be permitted, using fifty (50) fuze assemblies. If the total number of ailures obtained in the original test plus the retest equals or exceeds seven (7), the lot shall be rejected.

4.3.3.11 Transportation vibration (See 3.12) Critical Defect, Defect Code 19001.-A sample of one hundred fifty (150) fuze assemblies be selected from each lot and subjected to this test. The test shall be performed as specified in 4.4.12, using equipment in accordance with 4.3.4. If one (1) or more fuze assemblies fail to comply with the transportation vibration test requirement, the lot shall be rejected.

4.3.3.12 Functioning of fuze assembly (See 3.13) Defect Code 20001.-One hundred fifty (150) fuze assemblies subjected to the transportation vibration test in 4.3.3.11 shall be subjected to this test. The test shall be performed as specified in 4.4.13, using equipment in accordance with 4.3.4. If in the original test, fourteen (14) or more fuze assemblies fail to comply with the applicable requirement the lot shall be rejected. If six (6) to thirteen (13) fuze assemblies fail to comply with the requirements, a retest will be permitted, using three hundred (300) fuze assemblies. (These fuze assemblies must be first subjected to the transportation vibration test in 4.4.11, using equipment in accordance with 4.3.4. If one (1) or more fuze assemblies fail the transportation vibration test, the lot shall be rejected.) If the total number of failures obtained in the original plus retest equals or exceeds fourteen (14) the lot shall be rejected.

4.3.3.13 Check test for deterioration of detonator, stab, M55, Defect Code 21001.-If the total time between original acceptance of any detonator lot and the assembly of that lot into the fuze exceeds two (2) years, or if the detonators have been subjected to adverse conditions, however brief, at any time since previous tests, the detonator lot shall be subjected to and must satisfactorily pass the functioning test specified by the Government Inspector. This test shall be performed by the Government inspector at the facility assembling the detonators into the fuze.

4.3.4 Inspection equipment.-Equipment Tabulation (ET) number 9207971 and Equipment List (EL's) referenced thereon identify the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in 6.3.

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4.3.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of this specification may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey, and whenever inspection equipment drawings are needed.

4.3.4.2 Supply and maintenance.-Supply and maintenance of the equipment listed on the EL's shall be in accordance with MIL-I-45607.

4.3.4.3 Government use of contractor's inspection and test equipment.-The contractor shall, when required by the Government, make available all inspection and test equipment necessary for determining conformance with the contract requirements. Personnel for operating the equipment, and verification of its accuracy shall be supplied by the contractor for the performance of examination or test by the Government.

4.4 Test methods and procedures.

4.4.1 Locking of Detonator Rotor.-Prior to inserting the detonator, the housing assembly shall be probed to determine if the detonator rotor is in the fully armed position. If so, the assembly shall be classed defective and removed from the lot. If the detonator rotor is not fully armed, a machine probe inserted into the detonator hole of the detonator rotor will apply counter-clockwise pressure to assure that the rotor is locked in the unarmed position. Clockwise pressure will then be applied to move the detonator rotor as far as possible to the safe position. If the detonator rotor is found not to be locked in the unarmed position, the assembly shall be classified defective and removed from the lot (rotational directions are as viewed through the lead cup hole).

4.4.2 Salt spray of cover.-The plated covers shall be tested in accordance with Federal Specification QQ-Z-325 or QQ-P-416, as applicable. (Parts subjected to this test shall be discarded).

4.4.3 Arming of housing assembly.-The housing assembly, less detonator, shall be mounted in a fixture as shown on the Equipment List and rotated about its axis of symmetry to the arming speed specified. The direction of rotation shall be counter-clockwise, when looking at the blowout end. The angular acceleration shall not exceed that used in the non-arming test (4.4.4). The maximum time to reach the specified arming speed shall be six (6) seconds. Observation shall be made for arming, as evidenced by the detonator hole in the rotor being visible through the lead cup in the housing. (Parts subjected to this test shall be discarded).

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4.4.4 Non-arming of housing assembly.-The housing assembly, less detonator, shall be mounted in a fixture as shown on the Equipment List and rotated about its axis to the non-arming speed specified. The rotation must be in a counter-clockwise direction when looking at the blowout end. The angular acceleration shall not be less than that used in the arming test (4.4.3). Observation for non-arming shall be made, as evidenced by the detonator rotor being in the fully unarmed position.

"4.4.5 Verification of non-arming of housing assembly.-After completion of the non-arming test, the housing assembly, less detonator, shall be mounted in a fixture as shown on the Equipment List and rotated about its axis to the verification non-arming speed specified. The rotation must be in a counter-clockwise direction when looking at the blow-out end. The angular acceleration shall not be less than that used in the arming test (4.4.3). Observation for non-arming shall be made, as evidenced by the rotor being in the fully unarmed position."

4.4.6 Staking of firing pin assembly.-The firing pin assembly shall be assembled to the fixture listed on the Equipment List and the specified load shall be applied. (Assemblies subjected to this test shall be discarded).

4.4.7 Staking of detonator.-Before being subjected to this test, the detonator rotor shall be separated from the housing assembly and the rotor shaft removed from the rotor. The rotor shall then be assembled to the equipment listed on the Equipment List. The load specified on the applicable drawing shall then be applied. (Assemblies subjected to this test shall be discarded).

4.4.8 X-ray examination of fuze assemblies.-Armed or partial armed fuzes shall be determined by x-ray. Equipment and procedures shall be satisfactory to the Contracting Officer.

4.4.9 Jolt.-The fuze assemblies shall be assembled to the adapter listed on the Equipment List and subjected to the jolt test in accordance with MIL-STD-331. Arming of the fuze shall be determined by x-ray. The rotor must be locked in the unarmed position. (Assemblies subjected to this test shall be discarded).

4.4.10 Jumble.-The fuze assembly shall be subjected to the jumble test in accordance with MIL-STD-331. Arming of the fuze shall be determined by x-ray. The rotor must be locked in the unarmed position. (Assemblies subjected to this test shall be discarded).

4.4.11 Waterproofness.-The fuze assembly shall be immersed in water to a depth of not less than three (3) or more than eight (8) inches. The chamber shall be evacuated to a differential pressure

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of 11.0 ± 1.5 inches of mercury for a period of not less than one (1) minute. The water shall contain the wetting agenc Nacconal NR (Sodium Alkyl Aryl Sulfonate) or equal in a concentration of approximately 1 1/2 teaspoonful per gallon of water. (Assemblies subjected to this test shall be discarded).

4.4.12 Transportation vibration.-The fuze assemblies shall first be packed in a standard shipping carton. (Dummy fuzes shall be used to fill the remainder of carton). The carton shall then be securely fastened to the vibration machine and tested in accordance with MIL-STD-331. Arming of the fuze shall be determined by x-ray. The rotor must be locked in the unarmed position.

4.4.13 Functioning of fuze assembly.-After successful completion of the Transportation Vibration test, the fuze assembly shall be spun to 5000 ± 500 revolutions per minute (RPM). Spinning of the fuze assembly may be accomplished by use of test vehicle (Dwg. C9214362) and air jet or by use of spin test fixture (Dwg. F9202982). After the specified rate of spin is obtained, the assembly shall be allowed to drop 9 to 9.5 feet onto a 3/4 inch thick plywood sheet securely attached to a 1/2 inch thick plywood sheet. (The 3/4 inch sheet shall be replaced after a maximum of two hundred fifty (250) assemblies have been dropped on it). Observation shall be made as to whether the fuze fires on impact or in the air.

4.4.14 Safe position of Detonator Rotor.-The fuze assembly, just prior to installing the lead cup shall be probed to determine safe positioning of the detonator rotor. A machine probe which locates from the empty lead cup hole will determine the safe positioning of the detonator rotor. Any assembly which exhibits more than 15° of rotor movement from the locked position shall be classified defective and removed from the lot.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Level A - Military Package.-Not applicable

5.1.2 Level B - Limited Military Package.-Not applicable

5.1.3 Level C - Minimum Military Package.-Not applicable

5.2 Packing.

5.2.1 Level A - Military pack.-Not applicable

5.2.2 Level B - Limited military pack.-Not applicable

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5.2.3 Level C - Minimum military pack.-The fuzes shall be packed in accordance with Dwgs. 8882317 and 8882318.

5.3 Shipping.-When shipments of fuzes for more than one lot are made as a carload, each lot must be kept separate, and the division between lots clearly indicated, to prevent mixing in transit.

5.4 Marking.-Marking of the boxes shall be in accordance with Dwg. 8882318.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.
c. Provisions for submission of first article samples.

6.2 Inspection code numbers.-The five-digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment.-The contractor shall design inspection equipment as required by the referenced Equipment Lists (EL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

6.3.1 Equipment lists.-Equipment lists indicate the availability of inspection equipment designs by showing in the "Part Number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the "Part Number" column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 Equipment list codes.-The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.
MU - Army design, mandatory for use.

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6.3.3 Army designs.-Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications: Mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an equipment list references mandatory Army Designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, with the approval of the Government, design alternate inspection equipment to facilitate his operations. Such contractor-prepared designs (See 6.3.4) or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.4 Contractor designs.-Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finishes, notes and operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls:

All submitted contractor designs shall conform to MIL-D-1000, Category E, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high-contract reproductions. Contractors shall submit one copy of final designs as a rolled set. Designs shall be submitted for approval as specified in 6.3.5.

6.3.5 Submission of contractor design.-All submitted designs shall contain a reference to the applicable 5-digit code number contained in Section 4 of this specification and the appropriate component or assembly drawing number and revision letter to which the specific design applies. Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-QA-A-S. All other designs of inspection equipment shall be approved by the inspection element

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of the agency administering the contract. Submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished, normally, within one month after receipt.

6.4 Cost of check test-The Contracting Officer will arrange for the contractor to be reimbursed for the expense incurred in the performance of the check test for deterioration of the detonators. The tests shall be conducted at Government expense without cost to the coantractor who loaded the detonators or to the contractor assembling the detonators into the fuzes, and shall not constitute a basis for rejection against either contractor except where deteration has occurred as a direct result of carelessness in handling, storage, etc, permitted while the detonator lot was under the jurisdiction of either contractor.

6.5 Submission of test data.-In addition to the normal distribution of records, when the Fuze, Grenade, M219E1 is procured by the U.S. Army Munitions Command one (1) copy of all functioning data and ammunitin data cards shall be forwarded to: Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-QA-A-S.

6.6 Submission of results of contractor-conducted examinations and tests.-Unless otherwise specified by the Contracting Officer, the contractor shall forward requested records of examinations or tests to Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-QA-A-S.

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Preparing Activity:
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Project Number: 1330-A004

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